

Scalability & Future Plan

Date	04 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset size	May reduce prediction accuracy for unseen conditions	High
2	Single-user deployment	Cannot support many concurrent users	Medium
3	Basic security mechanisms	Risk of unauthorized access	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy on cloud with load balancing
2	Data Storage	Local dataset	Use cloud database and scalable storage
3	Performance	Basic ML prediction	Optimize model and enable caching
4	Security	Simple authentication	Add JWT, HTTPS and role-based access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Increase dataset and retrain ML model	1-2 Months	Improved prediction accuracy
Phase 2	Mobile application for farmers	3-4 Months	Better accessibility
Phase 3	IoT sensor integration	5-6 Months	Real-time recommendations
Phase 4	Satellite & weather API integration	6-12 Months	Highly accurate smart farming support